Hackathon Prompt:
The Quantum Safe

Background and motivation:

You have discovered a mysterious quantum safe.

The safe contains a secret code, but the only way to unlock it is by creating the correct quantum state.

Your mission is to reverse-engineer the quantum circuit.

Getting started:

You are given a target measurement distribution.

For example:

- 00 → 50%
- 11 → 50%

Your task is to create a quantum circuit that produces this distribution.

You may experiment with:

- H gates
- X gates
- Z gates
- CNOT gates
- Measurement

Your Mission:

Create a circuit that generates the required output.

The safe will "unlock" if your measured distribution is sufficiently close to the target.

Hint:

Think about what happens when you:

1. Put one qubit into superposition.
2. Connect it to another qubit.
3. Measure both.

What relationship do you observe between the two results?

Deeper Questions:

Why are 00 and 11 correlated?

What happens if you remove the CNOT gate?

What happens if you change which qubit receives the H gate?

Stretch Challenge:

Create three different circuits that produce the same measurement distribution.

Explain why they work.

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